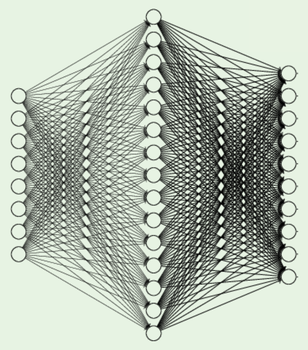
Nanophotonic Particle Simulation andInverse Design Using Artificial Neural Networks

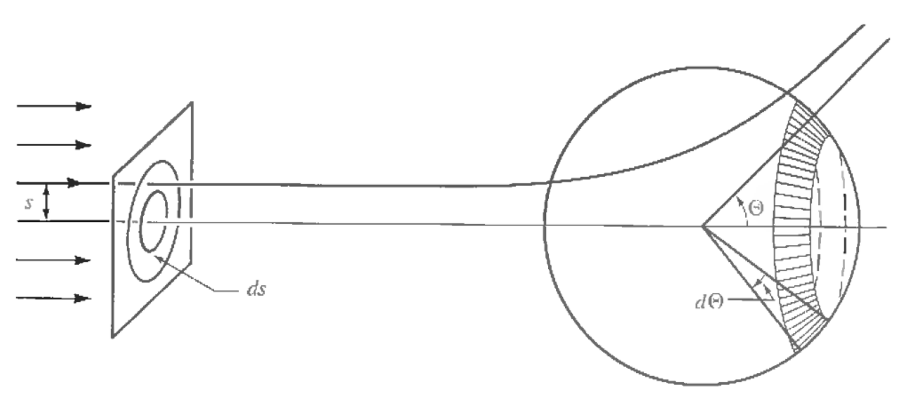


John Peurifoy, Yichen Shen, Li Jing, Yi Yang, Fidel Cano-Renteria, Brendan Delacy, Max Tegmark, John D. Joannopolous, and Marin Soljacic

Massachusetts Institute of Technology Department of Physics

NN's could learn Maxwell's Equations

- Inverse Design is difficult
  - You can compute the forward computation easily in many cases, but going the opposite direction is difficult
- NN's have an analytical gradient/matrix representation, which may be helpful to solve inverse design problems
  - Could Approximation simulations faster
  - Solve Inverse Design more accurately
  - Create an optimizer for desired results.



$$V(\vec{r}) = \frac{ZZ'ee'}{r}$$

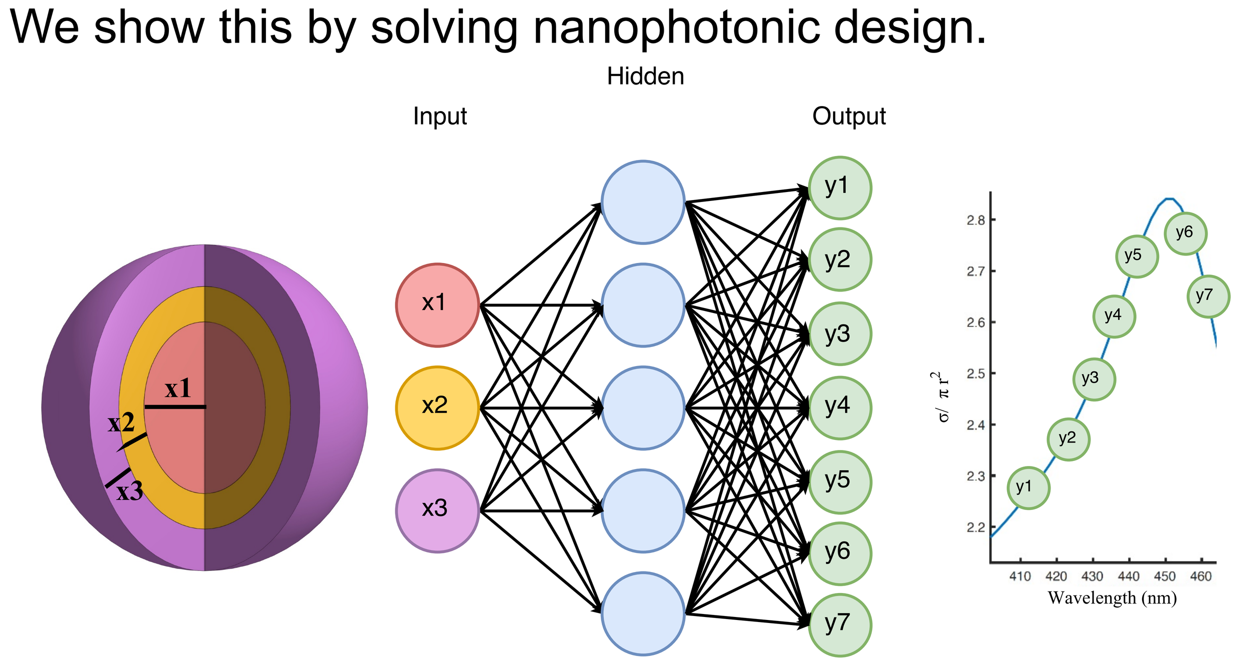
$$\left(-\frac{\hbar^2}{2\mu}\nabla^2 + \frac{ZZ'ee'}{r}\right)\psi(\vec{r}) = E\,\psi(\vec{r})$$

$$\psi(\vec{r}) = e^{ikz}\,g(r-z)$$

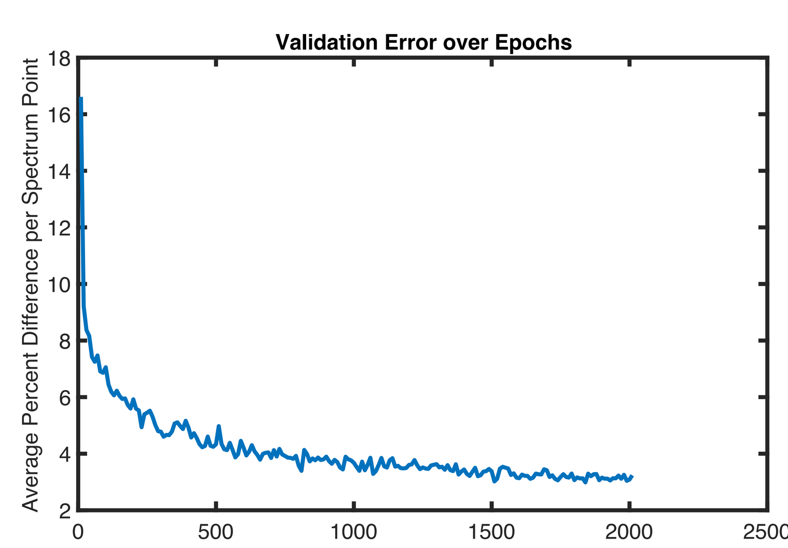
$$f(k,\theta) = -\gamma\frac{\Gamma(1+i\gamma)}{\Gamma(1-i\gamma)}\frac{e^{-i\gamma\log\sin^2\theta/2}}{2k\sin^2\theta/2} = -\frac{\gamma}{2k\sin^2\theta/2}e^{i(\eta_0-\gamma\log\sin^2\theta/2)}$$

$$\frac{d\sigma}{d\Omega} = \frac{\gamma^2}{4k^2\sin^4\theta/2}$$

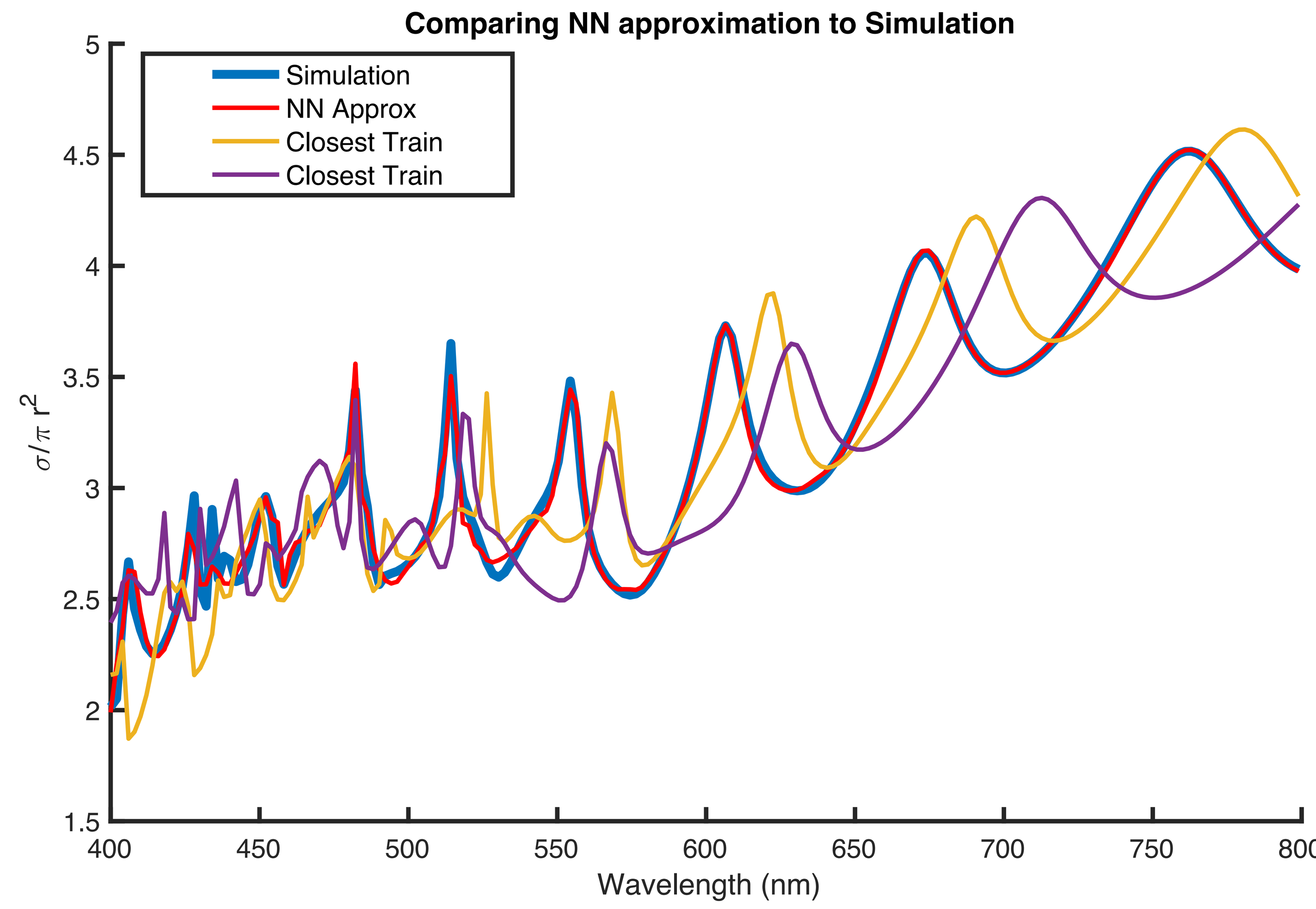
- We show this by solving nanophotonic design.



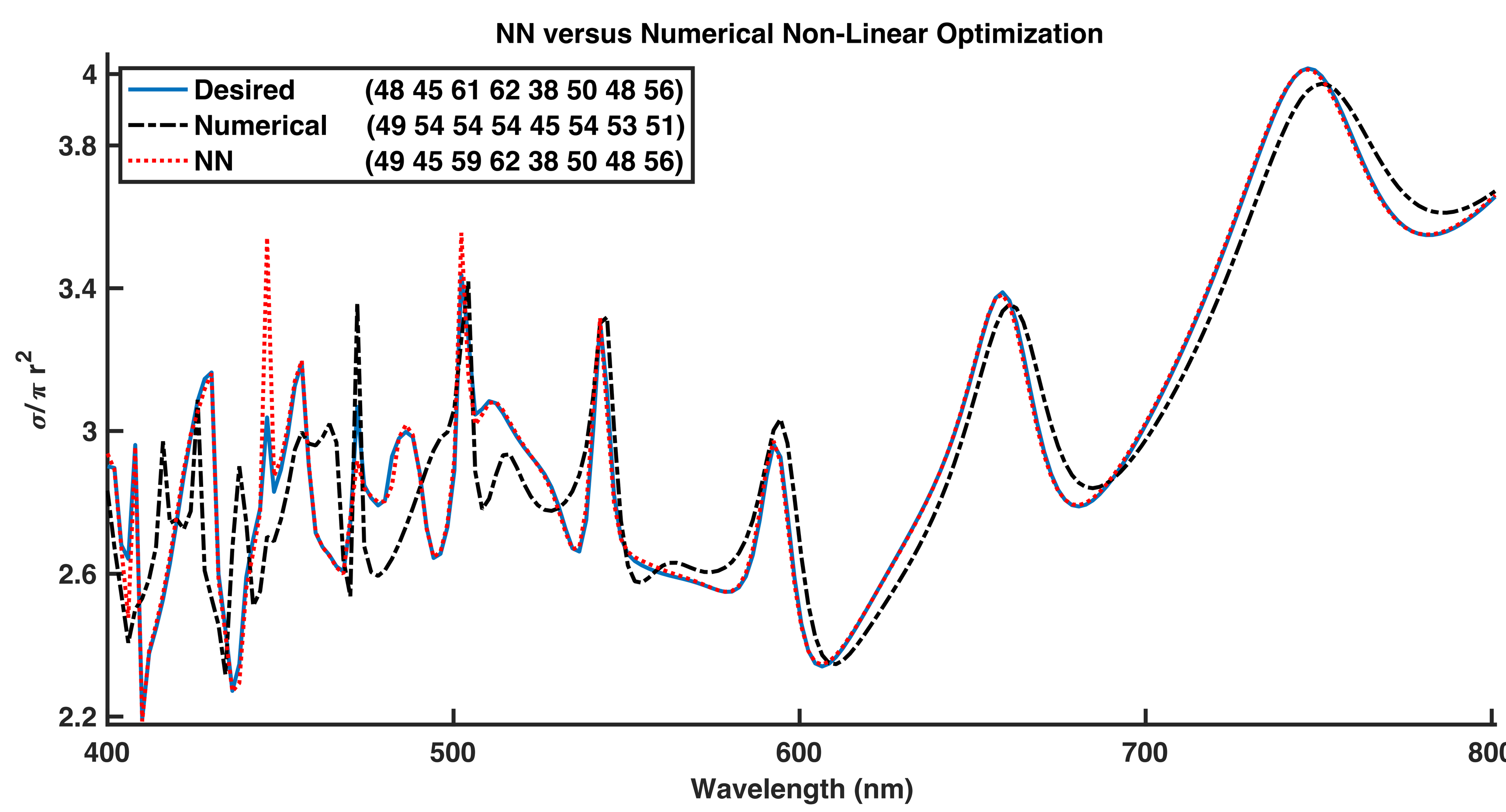
- The geometry (individual layer thicknesses) are input to the NN.
  - 8 layers , 30-70nm thick
  - Alternating TiO2/Silica
- A Matlab simulation generates the training data.
  - ~50,000 examples



NN's solve Nanophotonic Inverse Design



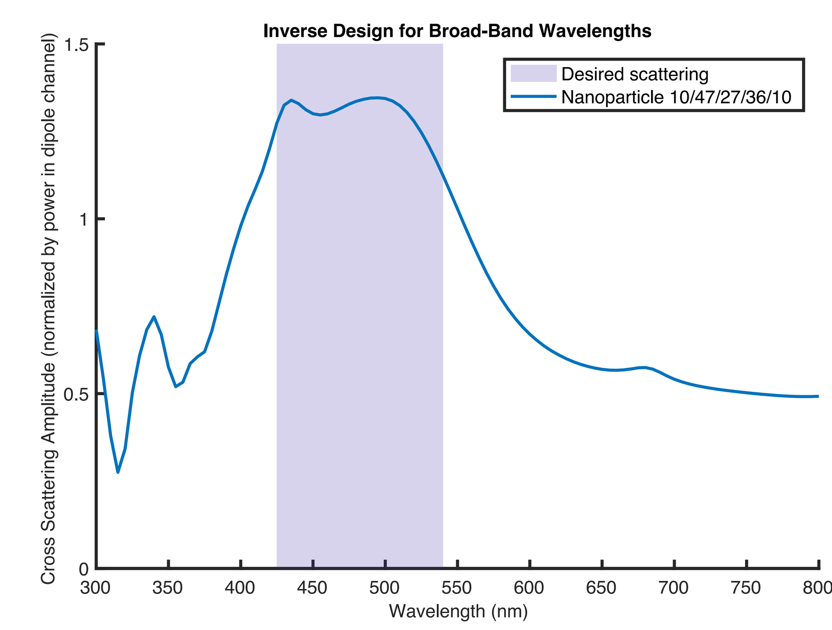
- The yellow/purple line show the nearest training examples in size.
- This had 8 input parameters, and was only shown 50,000 samples (~4 samples/input)



- To solve inverse design, we fix the weights and train the inputs as biases
- This result is for an eight layer nanoparticle.
- Blue is the desired spectrum, the black is from numerical non-linear optimizations, and the red is the Neural Network results.
  - This was compared against interior-point methods, initialized 50 times

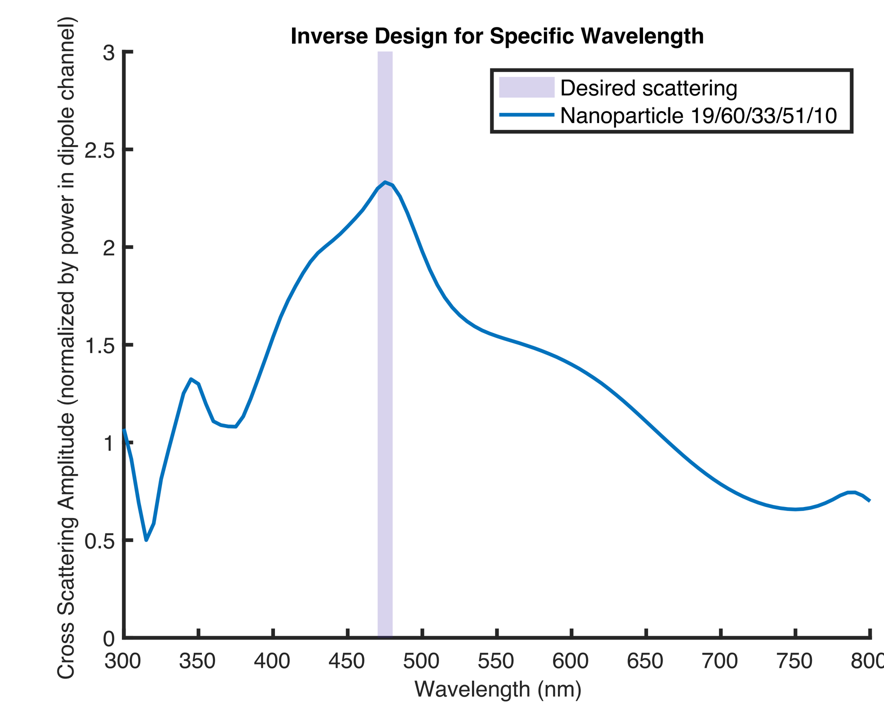
NN's can be an Optimization Tool

- We choose a loss function to optimize for a particular scattering

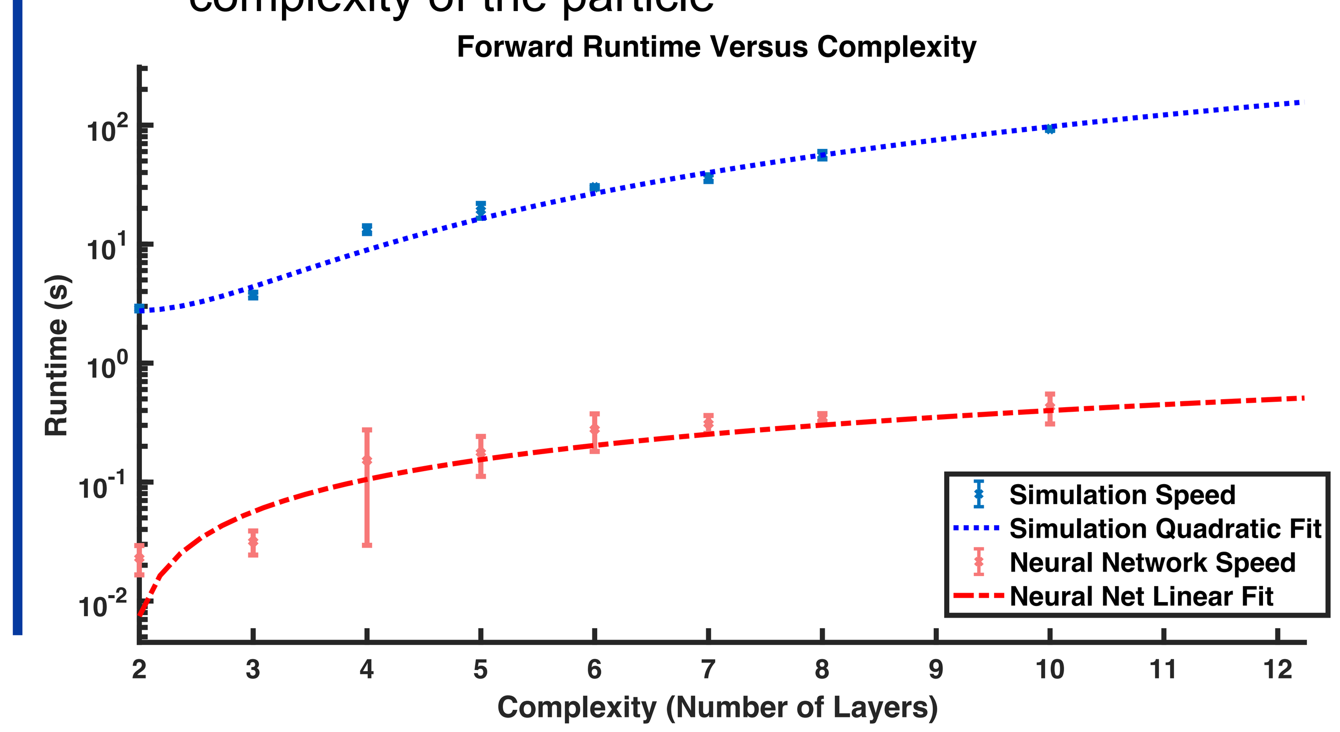


- The cost function here was maximizing purple region, minimizing outside.
- After a couple of iterations, it finds a broad-band material.

- Similarly, the procedure can be used for specific-wavelength scattering
- Applicable even if the underlying materials do not have sharp resonances

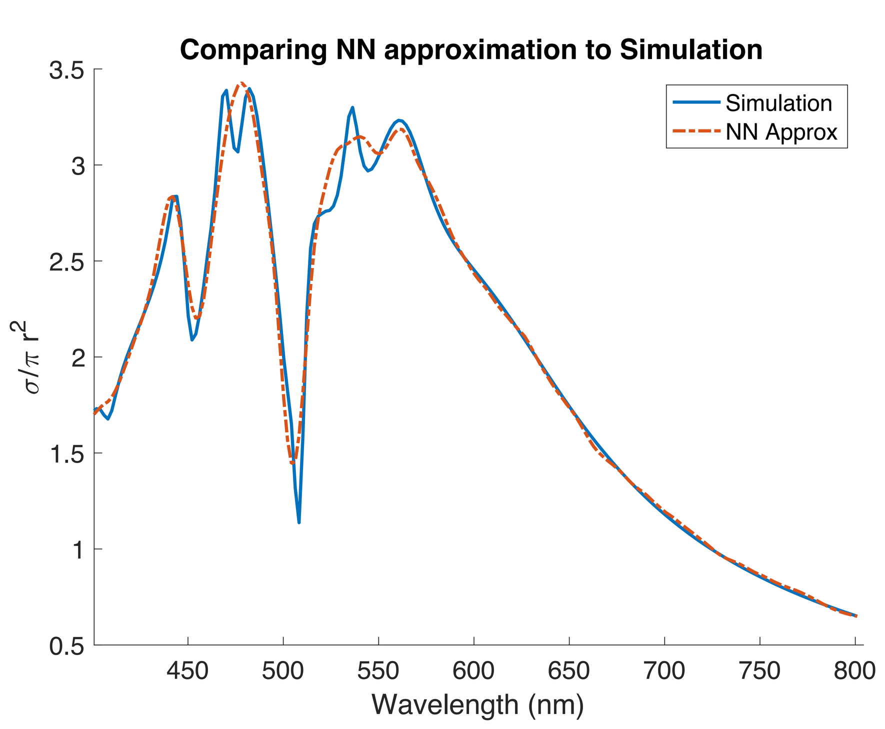


- Comparing the speed of the simulation/NN versus complexity of the particle



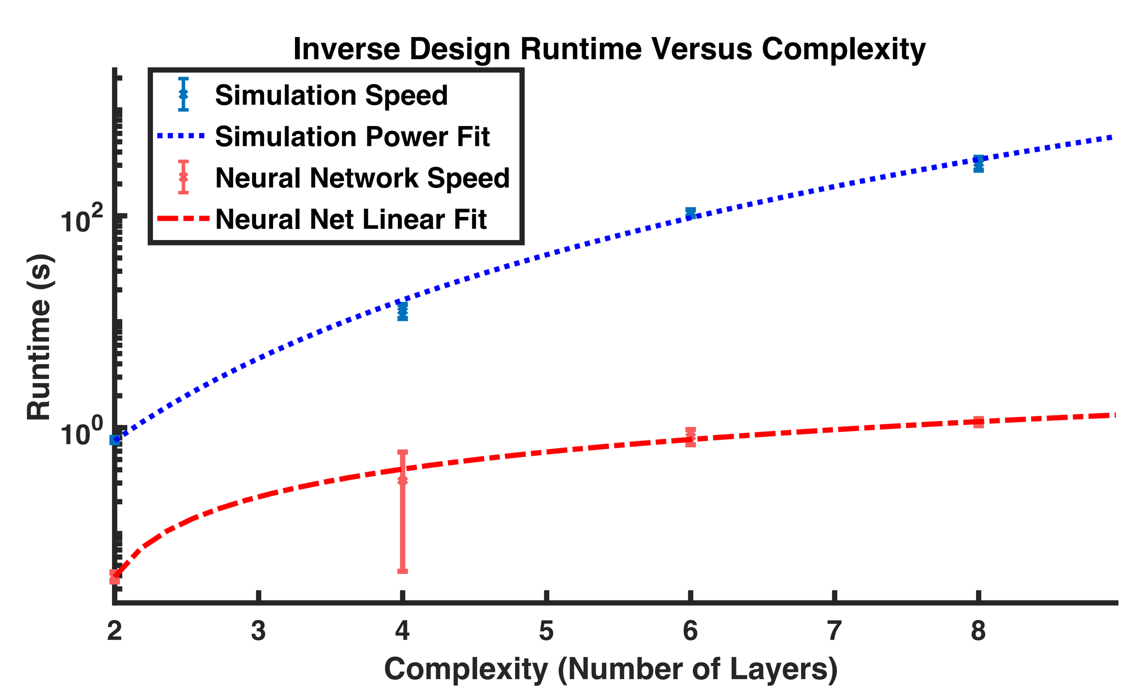
Exotic Materials: J-Aggregates

- Nanoparticles made with J-Aggregates have sharp resonances.
- Sharper resonances made the spectrums vary greatly
- NN still well-behaved.
  - Trained on same number of examples.
  - Only 3 layer particle
    - Metal, Dielectric, J-Agg



Inverse-Design Speeds

- Comparing time to reach a desired threshold



Conclusions

- Demonstrated NN can learn/approximate Maxwell interactions
- Showed NN's can solve Nanophotonic Inverse Design
- Constructed broad-band and specific-wavelength materials.
- Compared the speeds, and suggested that NN can open up previously insolveable inverse design problems.

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References

J. E. Peurifoy, Y. Shen, L. Jing, F. Cano-Renteria, Y. Yang, J. D. Joannopoulos, M. Tegmark, and M. Soljacic, in Frontiers in Optics 2017 (Optical Society of America, 2017) p. FTh4A.4.

ljing@mit.edu jpeurifo@mit.edu

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